

Provisional Patent Application prepared and filed by inventor (pro se).

Stardust to Sovereignty: Resonance-Based Knowledge Organization System (Specification)

TITLE OF THE INVENTION

Stardust to Sovereignty: Resonance-Based Knowledge Organization System.

FIELD OF THE INVENTION

The present invention relates to computer-implemented data-processing systems and methods that measure, model, and organize coherence across dynamic data fields using resonance-based computation. It further relates to artificial intelligence, computational epistemology, and information-organization architectures.

BACKGROUND OF THE INVENTION

Existing artificial-intelligence and knowledge-organization systems rely on optimization and probability models that evaluate information in oppositional or binary terms. Such systems fail to detect intrinsic alignment or coherence across data elements and often produce unstable or non-convergent results. There is a need for a computational method that detects and maintains coherence in digital information systems, providing self-stabilizing data organization and measurable alignment metrics.

SUMMARY OF THE INVENTION

The invention provides a computer-implemented system for organizing and processing information through resonance-based computation within a modular coherence topology. Digital input data are parsed into resonance domains, each represented by multi-dimensional coherence vectors. The system computes resonance metrics (R_{ij}) between domains, monitors convergence through a halting factor ($|dC/dt| < \epsilon$), and generates symbolic or visual outputs via a translation layer ($T(\Psi) \rightarrow S$). These processes transform unstructured data into coherence-ordered structures, improving computational stability, interpretability, and efficiency in artificial-intelligence and knowledge-systems applications.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates the system architecture of the resonance-based knowledge organization framework.

FIG. 2 depicts the resonance-computation process showing calculation of R_{ij} between domains.

FIG. 3 shows the halting-convergence graph where $dC/dt \rightarrow 0$.

FIG. 4 presents the field-propagation schematic of the information-coherence field.

FIG. 5 demonstrates the translation layer mapping $T(\Psi) \rightarrow S$ that converts resonance matrices to symbolic output.

DETAILED DESCRIPTION OF THE INVENTION

The system comprises five main computational modules implemented by one or more processors executing stored instructions on a non-transitory computer-readable medium:

Resonance Engine: Computes similarity metrics (R_{ij}) between resonance domains using cosine similarity, Jaccard overlap, and temporal decay weighting.

Calculus of Coherence: Validates internal coherence of computed domains.

Halting Factor: Detects convergence by monitoring the derivative of global coherence ($|dC/dt|$).

Field Computation Module: Propagates and stabilizes coherence fields using differential modeling ($\partial\Psi/\partial t = \nabla \cdot (\kappa \nabla \Psi) + R(\Psi)$).

Translation Layer: Generates symbolic or textual output from stabilized field states.

In one embodiment, the system was executed on a GPU-enabled computing device using real-time monitoring and matrix operations. Halting was detected at iteration step 13 when $|dC/dt| = 0.000460 < 0.001$, yielding a global coherence $C = 0.86$. Symbolic output described domain alignment as 'node1:● node2:●.'

CROSS-REFERENCE TO ANNEX A (VALIDATION FRAMEWORK)

The contents of Annex A – Validation Framework for the S2S System (Exhibits A–E) are incorporated herein by reference in their entirety for all purposes. Annex A provides experimental and computational validation supporting the embodiments described in this specification. Each validation module (Calculus of Coherence, Resonance Computation, Halting Factor, Field Computation, and Translation Layer) demonstrates measurable transformations of digital data performed by one or more processors. These validations further evidence that the disclosed system achieves a practical application of resonance-based coherence computation and substantiate enablement under 35 U.S.C. §112. These results demonstrate a concrete transformation of digital data structures and a measurable improvement in computational performance.

CROSS-REFERENCE TO ANNEX B (TECHNICAL APPENDIX)

The contents of Annex B – Technical Appendix for S2S System are incorporated herein by reference in their entirety for all purposes.

CROSS-REFERENCE TO ANNEX C (PRIOR ART ANALYSIS)

The contents of Annex C – Prior Art Analysis for S2S System are incorporated herein by reference in their entirety for all purposes.

CROSS-REFERENCE TO ANNEX D (CORE PIPELINE SPECIFICATION)

The contents of Annex D – Core Pipeline Specification for S2S System are incorporated herein by reference in their entirety for all purposes.

CLAIMS (PLACEHOLDER)

No claims are presented in this provisional application.

ABSTRACT

A computer-implemented system for measuring and organizing information through resonance-based computation. Input data are parsed into resonance domains, resonance metrics (R_{ij}) are computed between domains, convergence is detected when $|dC/dt| < \epsilon$, and a translation layer generates symbolic or visual outputs. The invention transforms unstructured data into coherence-ordered structures, providing improved computational stability and interpretability within artificial-intelligence and knowledge-organization systems.